

5.3.2 Areas Between Polar Curves

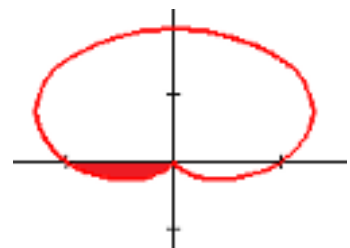
Motivation: Suppose we wish to find the area of some region bounded by curve(s) given in polar form.

Theorem: The area enclosed by the polar curve $r = f(\theta)$ for $a \leq \theta \leq b$ and the lines $\theta = a$ and $\theta = b$ is

Question to the Class: Does the above integral yield a signed area?

Example 1: Find the area enclosed by one of the loops of the four leaf rose given by $r = \cos(2\theta)$.

Example 2 (you try): Find the area that is shaded in the in the following graph of the curve $r = 1 + \sin(\theta)$.



Remark: As one would likely expect, if a region R is bounded by the curves with polar equations: $\theta = a$, $\theta = b$, $r = f(\theta)$, and $r = g(\theta)$ where $f(\theta) \geq g(\theta) \geq 0$ and $a \leq b \leq 2\pi + a$, then the area A of R is given by: $A = \frac{1}{2} \int_a^b \left([f(\theta)]^2 - [g(\theta)]^2 \right) d\theta$.

Question: Why do we need f and g to be nonnegative?

Example 3: Find the area of the region that lies inside the curve given by $r = 3\cos(\theta)$ and outside the cardioid given by $r = 1 + \cos(\theta)$.

Example 4 (you try):

Suppose the curve C is given by $r = 3 + 2\cos(\theta)$. Also suppose that the curve C' is given by $r = 3 + 2\sin(\theta)$. Find the area of the region that lies inside both of these curves.

Note: We must be careful when determining limits of integration. Certain polar curves, namely roses, can be graphed on different intervals of θ .

Example 5: Determine how long it takes to trace out the graphs of the roses. $r = \cos(2\theta)$ and $r = \cos(3\theta)$.

Example 6:

(a) Set up an integral to represent the area enclosed by the entire curve $r = \cos(3\theta)$

(b) Set up an integral to represent the area enclosed by the entire curve $r = \cos(2\theta)$

Note: This subtlety of limits is important when working with functions that use negative radii.